

# Salted

*SMART CITIES OF THE FUTURE* — WEATHER-BASED  
PREDICTION OF SIDEWALK SLIP RISK IN WINTER



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# — Problem

During winter in Canada, residents often lack accessible, street-level information about slippery and hazardous sidewalk conditions—particularly the risk of black ice. This information gap limits people’s ability to take preventive actions and increases the likelihood of slips, falls, and related injuries.

Our prototype focuses specifically on Vancouver’s weather and sidewalk data to address this challenge.

## ***Why this matters:***

- In BC, an older adult is hospitalized due to a fall every 30 minutes
- Approximately one older adult dies from a fall each day



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# Solution



Color-coded street-level  
slipperiness insights



Optimized routes to  
avoid slipping



Prevent falls by enabling  
early action



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## Datasets (Features) used

### ***Weather & Moisture Conditions***

These features capture whether conditions are favorable for black ice formation.

### ***Urban Microclimate & Humidity***

These features approximate how long moisture may persist on the sidewalk surface.

### ***Solar Exposure & Street Orientation***

These features reflect how quickly ice may melt during daylight hours.

### ***Infrastructure & Pavement Condition***

These features account for structural factors that increase freezing risk.



## **Black Ice Risk Score**

All features are combined into a composite sidewalk hazard indicator

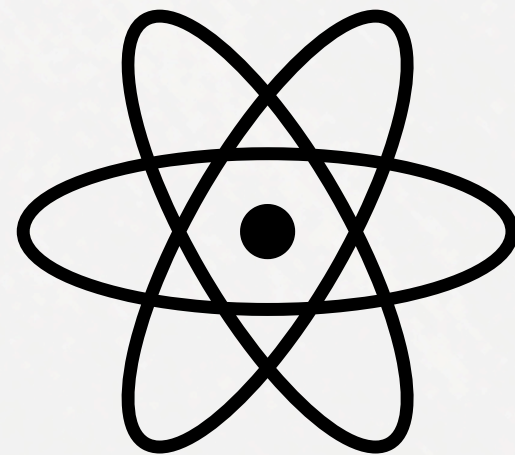


# Technical Architecture



## Data Sources (Inputs)

- OpenWeather API
- City of Vancouver Open Data of Pavement conditions



## Risk Scoring / Model Layer

Weighted composite scoring equation



## User-Facing Application Layer

Notification to let homeowners know when sidewalks might contain black ice so they can salt side walks



## Map

Allow users to plan the “least slippery” route to get from point A to point B based on black ice score and if the road has been salted



# Roadmap

## Phase 1

Adapting application to other weather events (i.e. rain during the fall season and heatwaves)

## Phase 3

Add alerts and personalization

## Phase 2

Partner with local municipalities to enforce regulations

## Phase 4

Embed AI to make better predictions